

OM ANANT WAIKAR

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Education

Binghamton University, State University of New York

Expected May 2027

Master of Science in Computer Science — GPA: 3.96/4.0

Binghamton, United States

Coursework: Design and Analysis of Algorithms, Systems Programming, Computer Networks, Cloud Computing

Indian Institute of Information Technology Vadodara

May 2025

B.Tech in Information Technology

Gujarat, India

Coursework: Operating Systems, Database Systems, Object-Oriented Programming (Java), Software Engineering

Technical Skills

Languages: Python, Dart, Java, C, JavaScript, SQL

Mobile & Backend: Flutter, FastAPI, REST APIs, Offline-First Storage

Databases & Caching: PostgreSQL, PostGIS, Redis, Realm

Cloud & Deployment: AWS EC2, Amazon Bedrock AgentCore, Docker

Algorithms: Graph Theory, Heuristic Algorithms, Constraint Modeling

Developer Tools: Git, GitHub, Postman, Pytest, Android Studio, Linux/Unix

Experience

Catalyze Systems Pvt Ltd

Jan 2025 – Jun 2025

Flutter Developer Intern

Remote

- Shipped **15+ Flutter/Dart screens**, including Add New Patient Visit, contributing to a **50–70% increase in active user engagement**
- Integrated **10+ RESTful APIs** to exchange patient data between the mobile frontend and backend services
- Designed and managed **15+ interrelated Realm database schemas** for offline-first patient-record storage
- Brought desktop **prescription management** to mobile, supporting custom medicine entries, dosage details, and on-device PDF generation

Indian Institute of Information Technology Vadodara

Jun 2024 – Jul 2024

Summer Research Intern

Gandhinagar, India

- Implemented **3 routing heuristics** based on distance and time-window opening/closing for electric-vehicle routing with time windows (EV-TSPTW) on a **6-node graph**
- Formulated a graph-based model with a travel-distance objective and **3 constraint categories** across a **15-edge network**
- Benchmarked **3 heuristics** on distance, waiting time, and feasibility: distance-based routing minimized travel, while time-window-closing achieved the highest feasibility at a higher distance cost

Projects

WaitWatch: Crowdsourced Wait-Time App | *Flutter, FastAPI, PostgreSQL, PostGIS, Redis, Docker, AWS*

- Built a Flutter app for reporting and viewing wait times across 5–8 dining locations, supported by **10–15 RESTful endpoints** built with FastAPI and PostgreSQL
- Used **PostGIS** for proximity validation and Redis caching to reduce repeated database queries by approximately 35%, with reports expiring after 45 minutes
- Deployed the backend on **AWS EC2** with Docker containers, maintaining 99%+ uptime across a two-week testing period

CloudOps Agent: Natural-Language Infrastructure Assistant | *Amazon Bedrock AgentCore, Python, AWS*

- Built an AgentCore agent that searches operational documents and selects tools for checking logs, deployments, and service health from natural-language requests
- Evaluated 50–75 requests with approximately 84% correct tool selection after refining **tool descriptions, input validation, and approval controls**
- Documented the agent architecture and tool-call workflows, enabling reuse across **3+ internal automation demos**

Leadership & Activities

Student Assistant, Office of International Students, Binghamton University

Feb 2026 – Present

- Supported **100+ international students weekly** through front-desk check-ins, inquiry handling, and referrals to university services
- Processed student e-forms and documentation in the **Sunapsis portal**, maintaining records and handling service requests